

```
import cv2
import matplotlib.pyplot as plt

# Load image
img = cv2.imread('image.avif')

# Convert BGR to RGB
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Create aliasing by shrinking and enlarging
# -----
small = cv2.resize(img, (100, 100), interpolation=cv2.INTER_NEAREST)
aliased = cv2.resize(small, (img.shape[1], img.shape[0]),
                     interpolation=cv2.INTER_NEAREST)

# -----
# Corrective approach:
# Apply Gaussian Blur (Anti-Aliasing)
# -----
blurred = cv2.GaussianBlur(img, (5, 5), 0)

# Resize using better interpolation
corrected = cv2.resize(blurred, (img.shape[1], img.shape[0]),
                      interpolation=cv2.INTER_CUBIC)

# -----
# Display Results
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(aliased)
plt.title("Aliased Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(corrected)
plt.title("Corrected (Anti-Aliased)")
plt.axis("off")

plt.tight_layout()
plt.show()
```

